

# AMARNATH S. PATEL

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## EDUCATION

### University of Central Florida

Undergraduate Student 4.00 GPA

B.S. in Physics (Optics & Lasers) and Mathematics, Computer Science Minor

August 2025 – Present

### Florida Atlantic University

3.66 GPA

Computer Science coursework concurrent with High School Diploma (111 Credit Hours)

August 2021 – May 2025

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## RESEARCH EXPERIENCE

### [UCF Astrophotonics Lab](#), CREOL – Undergraduate Researcher

August 2025 – Present

*Supervisor: Dr. Stephen Eikenberry*

- Developing instrumentation and automation software for astrophotonics research at the intersection of photonics and observational astronomy, spanning CREOL and the UCF Physics Department.
- Built an automated data acquisition and processing pipeline for modal characterization of photonic lanterns via off-axis digital holography; coordinates 4 instruments across 7-leg  $\times$  7-wavelength (1540–1570 nm) sweeps, implementing FFT-based twin-image extraction, Butterworth filtering, quadratic phase correction, and LP mode decomposition to recover complex modal amplitudes and field structure.
- Developing PolyOculus, control and automation software for an 8-telescope photometric observation array; implementing INDI-protocol mount control, focuser automation, RA/Dec coordinate slewing, and backlash compensation.
- Developed Python SDK wrapper (gentec-camera) for an IR beam profiling camera enabling automated acquisition, real-time beam analysis, and FITS output.

### UCF Physics Department – Paid Undergraduate Research Assistant

March 2026 – Present

*Physics Education Research – Supervisor: Dr. Zhongzhou Chen*

- Developing ESTELA, an automated multi-version exam generation system for AI-assisted isomorphic physics problem banks, supporting scalable and equitable assessment infrastructure for introductory STEM courses.
- Built in Rust with a vanilla JS frontend; parses YAML problem banks across 13 topic areas supporting 11 question types, renders LaTeX math via KaTeX, and exports multi-version exams as LaTeX or print-ready HTML/PDF.
- Funded by NSF award 2421299 and Gates Foundation INV-076932.

### [FAU Grant-Funded AI Safety Research Project](#) (High School)

January 2024 – March 2025

*Supervisor: Tucker Hindle, Florida Atlantic University*

- Benchmarked 10+ coherence and detection methods (BERT NSP, GPT-2 perplexity, RoBERTa, LSA, NLI, burstiness) across 3,125+ generated sequences to evaluate approaches for identifying AI-generated text.
  - Identified GPT-2 perplexity as the strongest discriminator ( $3.3\times$  gap between coherent and incoherent text); BERT NSP failed to distinguish the two.
  - Characterized a fundamental quality–evasion tradeoff across all humanization approaches tested, including iterative sampling, content abstraction, and RL-based methods.
  - Grant-funded with HPC access; findings presented at the Wilkes Honors College Symposium (2025).
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## OTHER EXPERIENCE

### Teaching Assistant – Calculus (High School)

August 2024 – May 2025

- Assisted 70 undergraduate students with learning calculus; office hours, exam review, grading. Part-time (10 h/week).

### [Advanced Experimental Vehicles](#) – Programmer, Leader, Builder (High School)

November 2023 – May 2025

- Configured Arch Linux ARM on Raspberry Pi 5 with Hyprland compositor and WireGuard VPN, enabling worldwide real-time telemetry monitoring of BMS, GPS, and camera feeds.
- Won 2nd Place in Division and Lockheed Martin Award for “Highest Level of Engineering Excellence.”

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## PRESENTATIONS

### Coherence and Detection Approaches for Identifying AI-Generated Text

2025

Wilkes Honors College Undergraduate Research Symposium, Florida Atlantic University.

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## SELECTED PROJECTS

### Photonic Lantern Digital Holography Automation

January 2026 – Present

- Automated data acquisition and processing pipeline for photonic lantern characterization via digital holography. Orchestrates tunable IR laser, fiber switch, motorized polarization controller, and GigE InGaAs camera across 7-leg  $\times$  7-wavelength sweeps (1540–1570 nm). Implements FFT-based fringe extraction and full 7-mode LP decomposition.

### [PolyOculus](#) — 8-Telescope Array Control Software

January 2026 – Present

- INDI-protocol mount control, automated focuser, RA/Dec slewing, and backlash compensation for an 8-telescope photometric observation array. Part of ongoing astrophotonics research at UCF CREOL.

### ESTELA — Problem Bank Visualizer & Exam Generator

March 2026 – Present

- Rust backend, vanilla JS frontend. Parses YAML problem banks across 13 topic areas and 11 question types. Renders LaTeX via KaTeX. Exports multi-version exams as LaTeX or print-ready HTML/PDF. NSF/Gates-funded research tool.

### [gentec-camera](#) — IR Beam Profiling SDK

November 2025

- Python SDK wrapper for Gentec IR beam profiling camera. Automated acquisition, real-time Gaussian beam analysis, FITS output for optical instrumentation research.
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## TECHNICAL SKILLS

**Programming Languages:** C/C++, Rust, Python, JavaScript, Shell (Fish, Bash, tcsh), LaTeX/Typst

**Tools & Libraries:** NumPy, SciPy, Docker, Git, CMake, XMake, INDI

**Instrumentation:** Tunable IR laser control, GigE camera acquisition, fiber switch automation, motorized polarization control, FITS data format

**Operating Systems:** Linux (Arch, NixOS, Ubuntu), BSD, Windows

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## RELEVANT COURSEWORK

### University of Central Florida

#### *Physics*

Geometric Optics & Lab (A)  
Chemistry Fundamentals I (A)  
Quantum Information Processing (A)  
Directed Independent Research (IP)

General Physics III (A)  
Chemistry Fundamentals II (IP)  
Theoretical Methods for Physics (A)  
Electricity & Magnetism I (IP)

#### *Mathematics*

Statistical Methods I (A)  
Linear Algebra (IP)  
Intro to Partial Differential Equations (IP)

Matrix & Linear Algebra (A)  
Introduction to Complex Variables (IP)  
Introduction to Graph Theory (IP)

#### *Computer Science*

Engineering Analysis & Computation (A)  
Object-Oriented Programming (IP)

Discrete Structures (A)

### Florida Atlantic University

#### *Physics*

General Physics I (C+)

General Physics II — Honors (A)

#### *Mathematics*

Calculus I (A-)  
Honors Calculus III (A)  
Matrix Theory (B)

Calculus II (A-)  
Differential Equations (A)

*Computer Science*

Data Structures & Algorithms (A)  
Computer Architecture (A-)  
Foundations of Computing (A)

Computer Logic Design (A)  
Intro to Deep Learning (A)

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**HONORS & AWARDS**

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|------------------------------------------------------------------------------------|------|
| Lockheed Martin Award — “Highest Level of Engineering Excellence,” AEV Competition | 2024 |
| 2nd Place in Division, Advanced Experimental Vehicles Competition                  | 2024 |
| NSF Award 2421299 — Research Funding (ESTELA project, with Dr. Zhongzhou Chen)     | 2026 |
| Gates Foundation INV-076932 — Research Funding (ESTELA project)                    | 2026 |